

Worksheet 5D: Trigonometric Identities and Equations — Solutions

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Fundamental Identities

1. 1
2. $\cos^2(x)$
3. $\sin^2(x)$
4. 1
5. $\cos(x)$
6. $1 + 2 \sin(x) \cos(x) = 1 + \sin(2x)$
7. $\tan(x)$
8. $\tan(x)$
9. $\cot(x)$
10. $\sin(x)$
11. $\tan(x)$
12. $\tan^2(x)$
13. $\tan^2(x)$
14. $\cot^2(x)$
15. 1
16. 1
17. $\sec^2(x) + \csc^2(x)$
18. $\sin(x)$

Part B: Sum and Difference

19. $\frac{\sqrt{6}+\sqrt{2}}{4}$
20. $\frac{\sqrt{6}-\sqrt{2}}{4}$

21. $\frac{\sqrt{6}-\sqrt{2}}{4}$
22. $\frac{\sqrt{6}+\sqrt{2}}{4}$
23. $2 + \sqrt{3}$
24. $\frac{\sqrt{6}+\sqrt{2}}{4}$
25. $\frac{\sqrt{2}-\sqrt{6}}{4}$
26. $A \sin(\omega t) + A \cos(\omega t)$
27. 0
28. (a) 0, 1 (b) $\frac{\sqrt{3}}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 0$
29. $\cos(105^\circ) = \frac{\sqrt{2}-\sqrt{6}}{4}$, $\sin(105^\circ) = \frac{\sqrt{6}+\sqrt{2}}{4}$

Part C: Double and Half Angle

30. $\frac{\sqrt{3}}{2}$
31. 0
32. $\frac{1}{2}$
33. $-\frac{1}{2}$
34. (a) $\frac{4}{5}$ (b) $\frac{24}{25}$ (c) $\frac{7}{25}$
35. $\sin(2\theta) = \frac{120}{169}$, $\cos(2\theta) = -\frac{119}{169}$
36. $\frac{\sqrt{2-\sqrt{2}}}{2}$
37. $\frac{\sqrt{2+\sqrt{2}}}{2}$
38. $\frac{\sqrt{2-\sqrt{3}}}{2}$
39. $\frac{\sqrt{2+\sqrt{3}}}{2}$
40. Power oscillates at twice the frequency of voltage
41. (a) $\phi = 0, 2\pi, 4\pi, \dots$ (b) $\phi = \pi, 3\pi, 5\pi, \dots$
42. 193 nm
43. $\theta \approx 19.7^\circ$

Part D: Solving Equations

44. $x = \frac{\pi}{6}, \frac{5\pi}{6}$
45. $x = \frac{3\pi}{4}, \frac{5\pi}{4}$
46. $x = \frac{\pi}{4}, \frac{5\pi}{4}$

47. $x = \frac{4\pi}{3}, \frac{5\pi}{3}$
48. $x = \frac{\pi}{3}, \frac{5\pi}{3}$
49. $x = \frac{4\pi}{3}, \frac{5\pi}{3}$
50. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
51. $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$ (Note: $\cos(x) = 0$ or $\cos(x) = 1$)
52. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$
53. $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$
54. $x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}$
55. $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$
56. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
57. $x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$
58. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
59. All t in $[0, 1)$ (they're always equal since frequencies match)
60. $x = 0, 0.5, 1.0, 1.5, 2.0$ meters
61. $\theta = 30^\circ, 90^\circ$ (for $m = 1, 2$; also negative angles)
62. $x = \frac{\pi}{6} + 2\pi n$ or $x = \frac{5\pi}{6} + 2\pi n$, where n is any integer
63. $x = \frac{\pi}{2} + \pi n$, where n is any integer
